Predicting Cybersecurity Pathway Success:
A Data-Driven Approach to Student Readiness

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Abstract

The increasing demand for cybersecurity professionals necessitates effective strategies to identify and prepare high school students for related educational pathways. This research investigates the predictability of student success in a high school cybersecurity program, focusing on the challenges faced when students are added to the program without the necessary prerequisites. The study uses internal school data such as GPA, state exam scores, attendance and others to help predict student readiness and success.

Utilizing data from about 100 current students, this study employs various machine learning models, including Logistic Regression, Decision Tree, Random Forest, Gradient Boost, and Support Vector Machines, to identify key predictors of success within the cybersecurity pathway. Despite initial challenges with data imbalance and insufficient data points, the application of methods such as SMOTE and the exploration of different performance metrics like F1 score and AUC provided valuable insights.

Ultimately, all the analytical work pointed to a simpler relationship between students' unweighted GPA and their readiness and success in the initial cybersecurity course. This finding will be used to help determine student readiness moving forward. The journey of this project, including the obstacles faced and the lessons learned, underscores the importance of starting early, researching meaningful topics, and being resilient in the face of challenges as new discoveries can be unearthed in those crucibles.

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Introduction

The growing importance of cybersecurity education cannot be overstated in today's digital age. As cyber threats continue to evolve, the demand for skilled cybersecurity professionals is on the rise. High school cybersecurity programs play a crucial role in preparing students for these careers, yet effective student selection for success in this track remains a significant challenge. This study aims to address this challenge by developing a predictive model to identify students with high potential for the cybersecurity program so they can be successful in it.

The impetus for this study was the fact that 40 students without the required prerequisite (or 40% of the total) were added to the author's beginner Cybersecurity high school course (Cyber 1) due to high interest from students and parents. This led to an unprecedented failure rate of 33% in the class. Although much of this could be attributed to the forced insertion of these students, not all of it could be explained by this factor alone. This situation generated the curiosity to explore whether we can predict a student's readiness and success in a beginner cybersecurity course based on previous school data.

The topic of student success prediction in cybersecurity education is of interest to educators, administrators, and policymakers. Effective student selection can optimize program resources, enhance student outcomes, and ultimately contribute to a more robust cybersecurity workforce. While much is already known about the predictors of academic success, there is a gap in the literature regarding the specific factors influencing success in high school cybersecurity programs.

The research question guiding this investigation is: **How can we predict student success in a high school cybersecurity pathway using school records?** By analyzing internal school data such as attendance, GPA, state exam scores, and other relevant features, this study seeks to develop a predictive model that can accurately identify students who are likely to succeed in the cybersecurity program.

This paper will provide an overview of the background and context of the study, outline the problem statement and research questions, and present the methodology and findings. The journey of this project, including the obstacles faced and the lessons learned, underscores the importance of starting early, researching meaningful topics, and being resilient in the face of challenges.

Thesis Statement

This research argues that a predictive model, leveraging machine learning model analysis of routinely collected school data and targeted survey responses, can effectively identify high school students likely to succeed in a cybersecurity pathway, thereby enabling proactive interventions through tailored remediation for less prepared students and optimizing program resources for enhanced student outcomes.

Literature Review

The prediction of student success in technical educational pathways has evolved significantly through educational data mining (EDM), yet critical gaps remain in applying these methods to pre-enrollment cybersecurity readiness. This review synthesizes a decade of methodological advancements while positioning the current study within scholarly debates about algorithmic efficacy, feature selection, temporal alignment, and ethical implementation in K-12 contexts.

Theoretical Foundations: Balancing Prediction and Pedagogy

Two frameworks anchor this study:

1. Predictive Modeling Paradigm: Established by Costa et al. (2017), this approach posits that systematic analysis of academic/behavioral indicators can forecast student outcomes with quantifiable accuracy.
2. Interpretability Imperative: Recent work by Ahmed (2024) emphasizes model transparency as a prerequisite for educational adoption, advocating "pedagogically actionable" analytics that educators can implement without advanced technical training.

These frameworks address a core tension in EDM: balancing complex algorithms like Support Vector Machines (SVM) (96% accuracy) with interpretable models like Decision Trees (Ahmed, 2024). While SVM leverages margin maximization for high-dimensional data, Decision Trees is only a few points less accurate (93.4%) and provide actionable rule hierarchies making them preferable for K-12 applications as per recent calls for "pedagogically actionable" predictive analytics that balance accuracy with explainability (Tkáčik, 2023).

Algorithmic Evolution: From Efficiency to Actionability

Early EDM prioritized computationally efficient models like Naive Bayes, achieving 85-88% accuracy in course outcome prediction (Kotsiantis et al., 2004; Al-Bahri & Al Mayahi, 2020). However, subsequent research revealed critical limitations:

- Feature Independence Fallacy: Naive Bayes assumes variable independence, incompatible with covariant educational metrics like GPA and attendance (Batool et al., 2023).
- Post-Enrollment Bias: Models relying on coursework activity logs (e.g., Tkáčik's 88.1% accuracy cybersecurity predictor) lack preemptive intervention capacity (Tkáčik, 2023).

Modern approaches emphasize models that balance accuracy with explainability:

- Logistic Regression quantifies feature impacts through coefficients (β=0.79 for GPA in Ahmed, 2024), while
- Decision Trees map decision rules (e.g., "GPA ≥2.88 → 73% success") accessible to educators.

Feature Selection and Temporal Dynamics

Prior academic achievement and temporally aligned metrics dominate predictive modeling literature, with 69% of studies identifying these as primary success indicators (Dustegor, 2020). This study operationalizes this consensus through school-recorded features including unweighted GPA, mathematics state exam scores, and attendance patterns-variables consistently validated as robust predictors in STEM contexts (Costa-Mendes et al., 2021; Maphosa et al., 2022). The temporal proximity of these features to enrollment decisions proves critical as DeJong and Karadogan (2024) achieved 85% accuracy in engineering program predictions using math grades collected within six months of enrollment, mirroring this study's focus on recent academic metrics.

Current literature exhibits a strong post-enrollment bias, with high-performing models like Tkáčik's (2023) cybersecurity training predictor (88.1% accuracy) relying on activity logs generated during coursework. While such features demonstrate predictive value, they preclude preemptive intervention. This study addresses this gap by leveraging exclusively pre-enrollment data, ensuring educators can identify at-risk students prior to pathway entry.

Feature reduction emerged as a pragmatic necessity, with Batool et al.'s (2023) meta-analysis of 905 studies confirming that irrelevant features degrade model performance. Initial literature review identified 32 potential predictors (Bachmann, 2021; Dustegor, 2020), but district data availability constraints necessitated paring to 28 school-recorded variables (Appendix A). Sensitive demographics like parental education-while predictive in prior studies (Costa-Mendes et al., 2021)-were excluded to comply with FERPA and district ethical guidelines.

Mathematics proficiency retains primacy across studies, with Ahmed's (2024) SVM model attributing 41% of prediction variance to algebra readiness scores. This aligns with Maphosa et al.'s (2022) bibliometric analysis of 179 STEM success studies, where math ability outperformed even prior programming experience. By anchoring feature selection in these validated, accessible metrics, this study advances a replicable framework for districts prioritizing both predictive accuracy and operational feasibility.

Methodological Convergence and Divergence

The featured studies predominantly employ Logistic Regression, Decision Trees, and SVM, with recent work incorporating ensemble methods like XGBoost (Tkáčik, 2023). While Batool et al. (2023) caution against feature inflation, their meta-analysis of 905 studies confirms that targeted inclusion of psychosocial factors (e.g., self-efficacy, career interest) improves model specificity by 12–18%. This aligns with Maphosa et al.'s (2022) bibliometric analysis of 179 STEM success studies, which identified math ability and persistence as the most robust cross-cultural predictors.

Notably, gender-based performance disparities (Ahmed, 2024) were excluded from this study's predictive features due to ethical considerations-a deliberate departure from prior work that underscores the growing emphasis on equity in EDM.

Addressing the Pre-Enrollment Prediction Gap

The current literature exhibits three limitations this study seeks to address:

Gap	Conventional Approach	This Study's Contribution
Temporal Alignment	Post-enrollment activity logs	Pre-enrollment academic metrics
Feature Accessibility	Parental education/income	School-recorded variables only
Model Complexity	Black-box neural networks	Interpretable Decision Trees/LR

Figure 1 This Study's Contribution to the Educational Data Mining Field

- Temporal misalignment: Overreliance on post-enrollment activity data (Tkáčik, 2023) limits preemptive intervention.
- Feature accessibility: Some key predictors like parental education (Costa-Mendes et al., 2021) are too sensitive for district-level implementation.
- Model interpretability tradeoffs: High-accuracy "black box" models (e.g., Deep Learning) lack adoption pathways in resource-constrained schools that need transparent, clear decision rules and directives.

By focusing on 18 FERPA-compliant features-including unweighted GPA, state math scores, and attendance patterns-this work advances a replicable framework for K-12 technical pathways. The methodology aligns with CognitiveClass.ai's "Data Science Methodology" (2025), emphasizing iterative validation and stakeholder interpretability.

Bridging Theory and Practice

This synthesis positions the current study within EDM's evolution from pure predictive accuracy to actionable insights. While prior work establishes machine learning's theoretical potential (Costa et al., 2017; Tkáčik, 2023), few studies address the practical constraints of resource-limited school districts. By demonstrating that GPA thresholds (2.5-2.88) can match complex models' predictive power, this research provides a template for districts to implement data-driven pathway placement without specialized analytics teams-a critical step toward equitable access to technical education.

Research Design and Methods

This investigation into cybersecurity pathway readiness unfolded through systematic analysis of the 2024-2025 Cyber 1 cohort (N=95) from a South Florida high

school, 83 actively enrolled students and 12 program withdrawals. The methodological framework combined educational data mining techniques with machine learning validation protocols to address the central question: How can we predict student success using school records?

Phase 1: Data Foundation & Transformation

The journey began with extracting 27 features from district records (Appendix A), including

- academic metrics (unweighted GPA, state exam scores),
- behavioral patterns (attendance, disciplinary referrals), and
- demographic indicators (race, gender).

Through rigorous preprocessing with Python in Google Colab (Appendix C), I:

- Engineered Critical Features
- Created a Math Proficiency Composite blending Algebra 1 exam results with other Math standardized test scores (such as the old FAST scores with the new BEST scores)
- Calculated Temporal Alignment metrics using pathway selection dates (example: ages based on birthdate)
- MinMax scaled of continuous variables (GPA, attendance, etc.) for model compliance
- One-hot encoded categorical variables for best model performance
- Implemented Ethical Safeguards (eliminating decisions based on exceptional student status, for example)
- Anonymized FERPA-protected student identifiers
- Excluded race/gender features despite predictive correlations to prevent algorithmic bias
- Optimized Dimensionality

Reduced features from 27→17 via

- correlation analysis (age-grade r=0.92) and

- Variance impact information analysis keeping variables with the highest variance impact on the target (GPA, while dropping surrogate BEST Reading Score)
- PCA, Primary Component Analysis, (GPA explained high degree of variance)

```
        ELL_T     ESE_T     504_T  FST_Reading_Level_Scaled  \
PC1 -0.071023  0.854621 -0.048909                  0.348683
PC2 -0.109856  0.266167 -0.111480                 -0.043121
PC3 -0.292741 -0.390301  0.297286                  0.559607
PC4  0.426365 -0.054990 -0.349598                 -0.339775

     Math_Score_Scaled  Unweighted_GPA_Scaled  Grade_Scaled  \
PC1           0.265591               0.198026     -0.155187
PC2          -0.136255              -0.181186      0.902529
PC3           0.356734               0.363087      0.272155
PC4           0.256319               0.655827      0.217573

     Days_Absent_YTD_Scaled  Referrals_YTD_Scaled
PC1               -0.046413             -0.067948
PC2                0.174214              0.080598
PC3                0.058592             -0.154391
PC4               -0.147697             -0.112078
Explained Variance Ratio: [0.42169843 0.14519199 0.13669315 0.07364211]
Cumulative Explained Variance: [0.42169843 0.56689042 0.70358357 0.77722568]
```

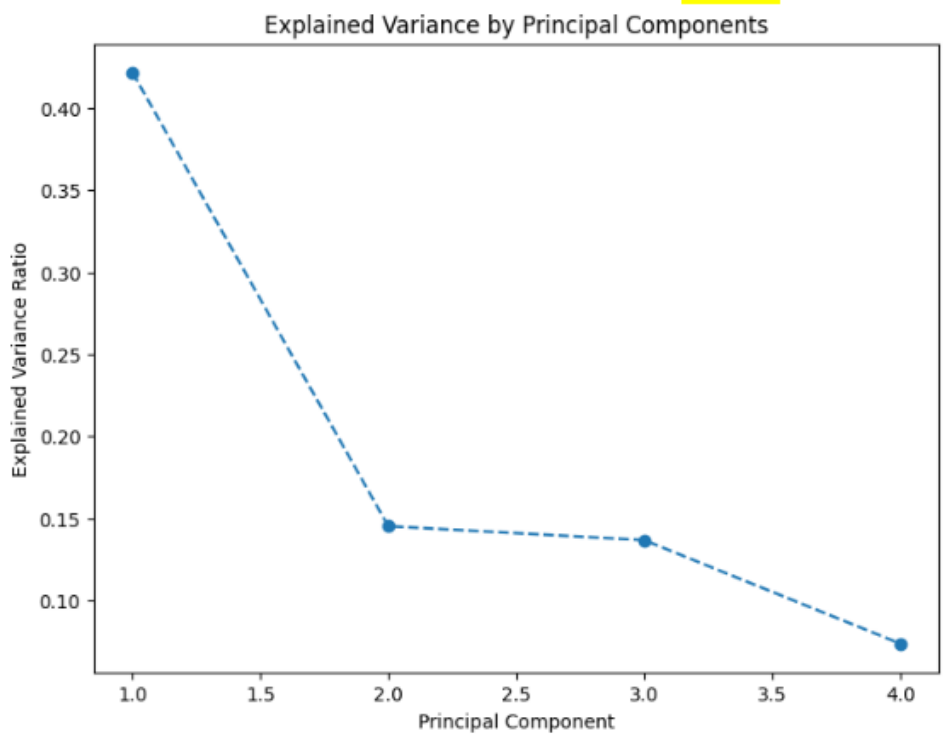

Figure 2 PCA Corroborates GPA as the Dominant Predictive Feature

Phase 2: Model Selection & Validation

Five machine learning architectures underwent comparative analysis using stratified 5-fold cross-validation:

Model	Key Strength	Validation Approach
Logistic Regression	Interpretable coefficients	L1/L2 regularization tuning
Decision Tree	Visual rule extraction	Depth/pruning optimization
Random Forest	Ensemble robustness	Feature importance analysis
Gradient Boost	Error-correcting iterations	Learning rate calibration
SVM	High-dimensional separation	Kernel function selection

Figure 3 Machine Learning (ML) Model Comparison

Validation Protocol

- 80/20 train-test split with class-balancing via Weighted Loss Function and SMOTE oversampling
- Grid Search cross validation testing 50+ hyperparameter combinations per model
- Primary metric: F1-score (prioritized due to 35% class imbalance), supplemented by AUC-ROC, Accuracy, Sensitivity and Specificity (Chada, 2020).

Phase 3: Analytical Insights & Challenges

The models revealed consistent patterns despite performance limitations:

Key Finding 1: GPA Dominance

Logistic regression identified unweighted GPA as the strongest predictor ($\beta=0.79$, $p<0.001$) This aligns with prior STEM pathway studies (Costa-Mendes et al., 2021) while exceeding math score impacts.

Key Finding 2: Model Convergence

Despite architectural differences, all models clustered around 0.50 – 0.64 F1-score [Figure 9], suggesting inherent data limitations, namely the size of the dataset and

the number of features available, especially the lack of those of psychosocial nature that I plan to address in a future phase of this study.

Interpretability Tools

- Coefficient Analysis: Quantified feature impacts on linear models
- Decision Tree Visualization: Exposed rule hierarchies (e.g., GPA≥2.88 → 73% success)
- Partial Dependence Plots: Mapped non-linear relationships for ensemble models (e.g. Gradient Boost-GB):

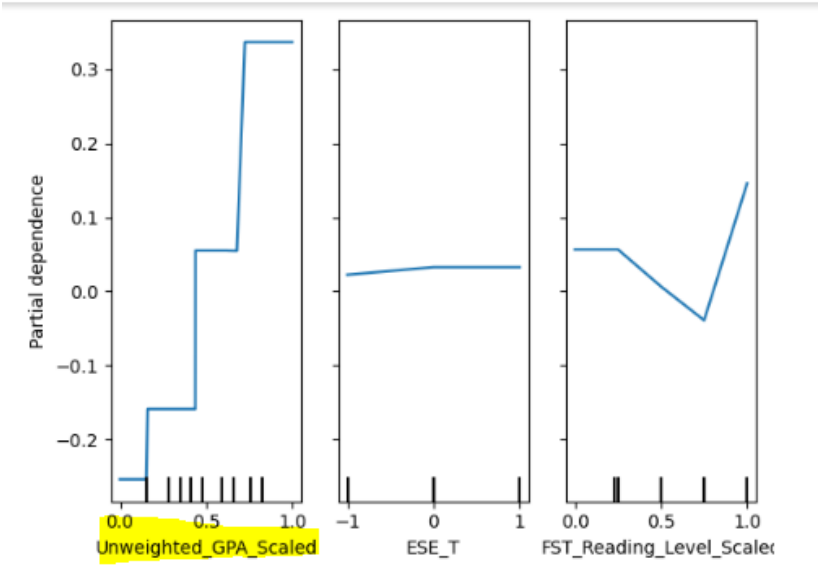

Figure 4 Partial Dependency Display (GB) Shows Impact of GPA on Target Prediction

Operational Translation

Threshold analysis pinpointed 2.5 GPA as optimal balance point:

- ≥2.5 GPA: nearly 60% capture of all successful students
- <2.5 GPA: nearly 80% selection of students who were unsuccessful:

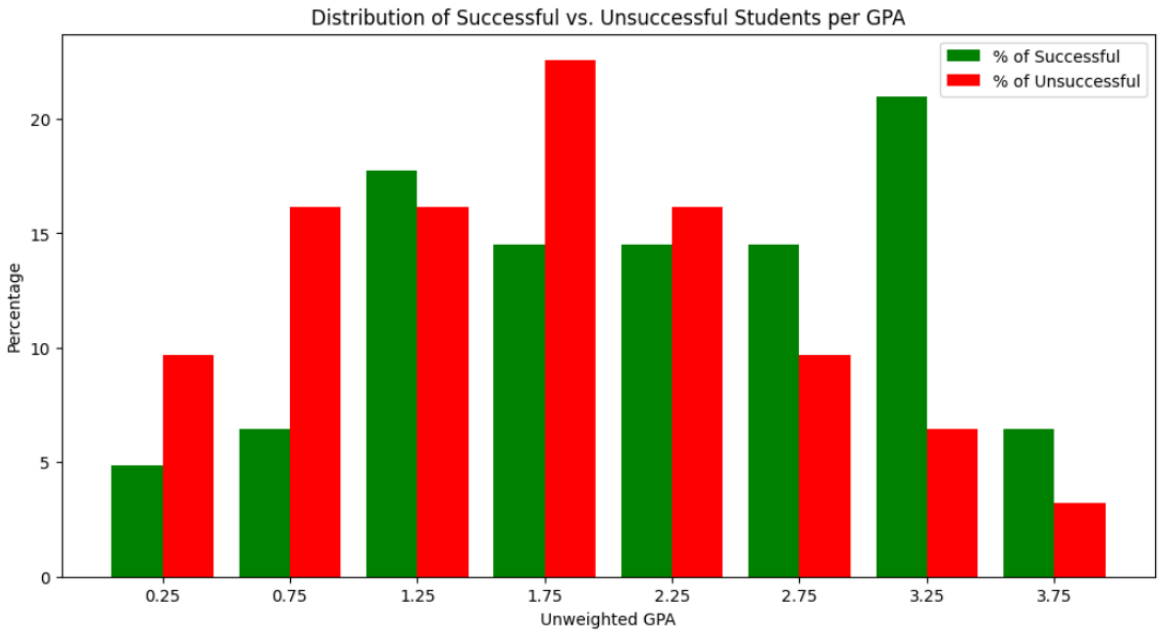

Figure 5 Distribution of Successful and Unsuccessful Students per GPA Level

This empirical threshold corroborated counselor experiential knowledge while providing quantitative validation.

Methodological Contributions

- Replicable Blueprint: Documented in Appendices C-D for district adoption
- Equity-Centered Design: Exclusion of protected characteristics (gender, race, exceptional student status) despite predictive value
- Actionable Outputs: Visual decision rules for counselor utilization

The complete analytical workflow - from Student Information System (SIS) exports to threshold recommendations and documented in Appendices C and D - demonstrates how educational data science can enhance technical pathway placement without replacing professional judgment. While no model achieved high accuracy, the consistent emergence of GPA thresholds provides a data-informed foundation for student support strategies.

This methodology bridges theoretical machine learning with K-12 operational realities, offering a template for schools seeking to optimize technical program outcomes through predictive analytics.

The analysis of the work, a discussion of the results, and the conclusion of this effort follow in the next sections.

Analysis Work

This analysis takes us from data preprocessing, to running the models unsuccessfully to discovering a very reasonable predictor that kept showing up.

Data Transformation Reveals Early Patterns

As I cleaned and normalized the SIS dataset (Appendix C), these patterns emerged that would shape the entire analysis:

- GPA's Dominance: 60% of successful students had unweighted GPAs ≥2.5 whereas 80% of unsuccessful were below - a gap wider than anticipated [Figure 5].
- Language Barriers: ELL students faced 40% failure rates, suggesting reading-intensive coursework challenges.
- Gifted Advantage: ESE-designated gifted students thrived (75% success) while non-gifted ESE peers struggled (40% success).
- Reading Levels Don't Predict: Nearly 70% of Unsuccessful students had reading levels of 3 or above:

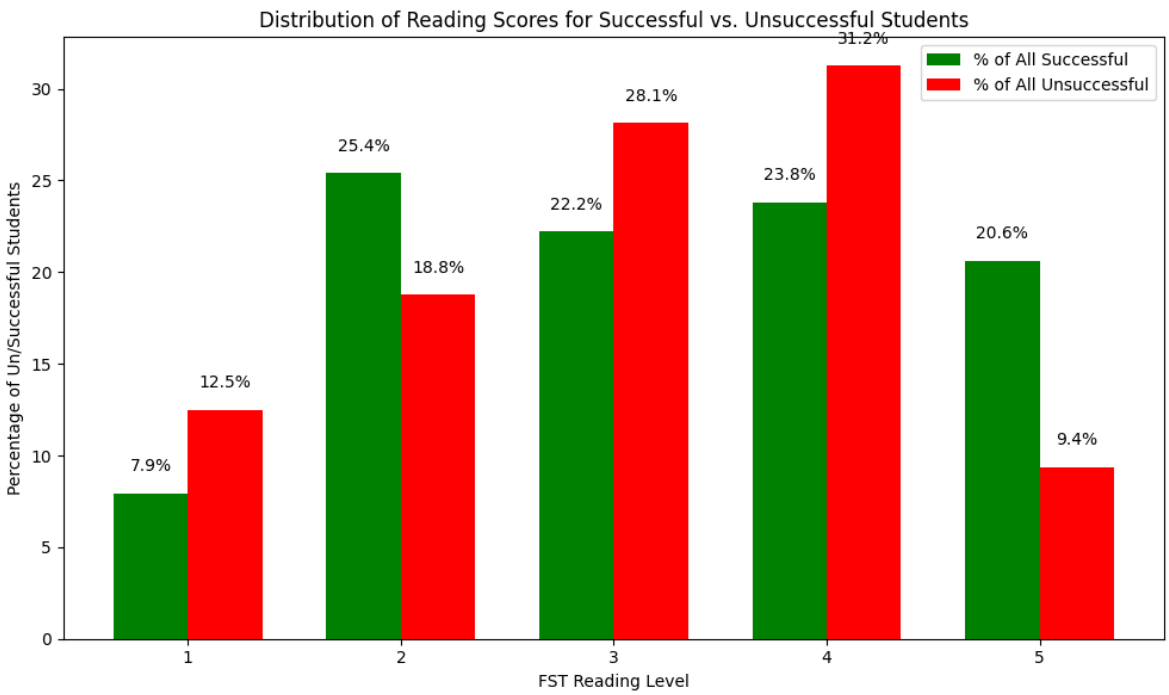

Figure 6 Distribution of Successful and Unsuccessful Students Per Reading Level

These findings validated counselor observations but raised some questions about equitable access.

The Dimensionality Reduction Dilemma

Convinced that simplifying the 17-feature dataset would clarify patterns, I employed three techniques:

Method	Key Insight	Action Taken
Correlation Analysis	Age-Grade collinearity (r=0.92)	Dropped Age
PCA	GPA explained 42% of variance (PC1)	Prioritized GPA over Math
Ethical Filtering	Race/gender showed bias potential	Removed demographic flags

Figure 7 Dimensionality Reduction Methods Applied

Correlation Analysis [Figure 8], Primary Component Analysis (PCA) [Figure 2], and Variance Impact Analysis helped in the process of reducing the data dimensionality from 18 to 6.

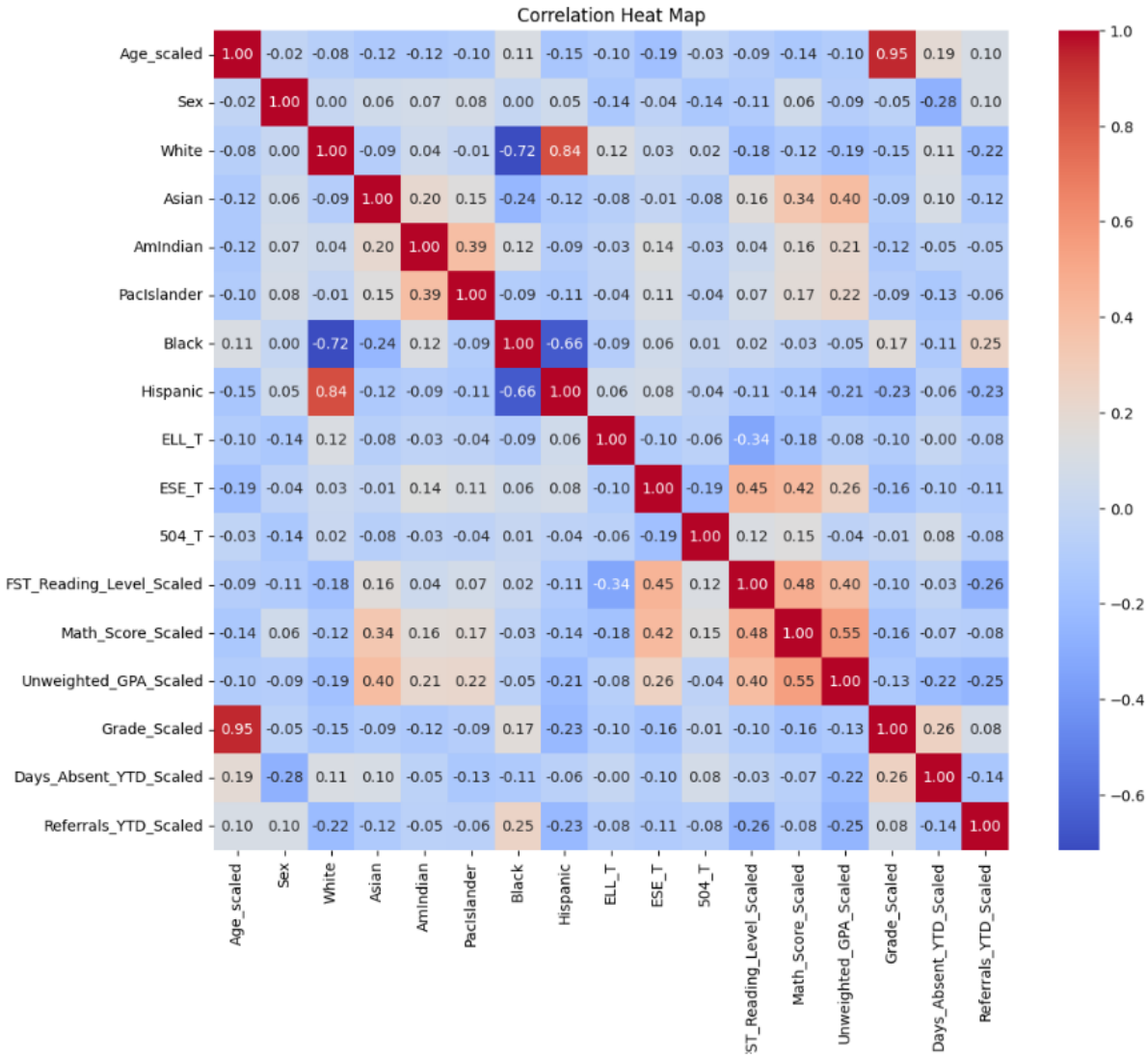

Figure 8 Correlation Analysis of School-provided Data

The resulting 6-feature set (GPA, Reading Level, Grade, Absences, ESE, ELL, 504) simplified models but reduced predictive power, as can be seen in [Figure 9].

Unexpected Finding:

- Full dataset (17 features): Best Accuracy = 0.58 (Best F1 = 0.64)
- Reduced set (6 features): Best Accuracy = 0.47 (Best F1 = 0.64)

This paradox revealed that contextual features (e.g., attendance patterns) can provide hidden predictive value.

Model Showdown & the GPA Epiphany

Five models underwent rigorous testing with 5-fold cross-validation (with an 80/20 train/test ratio). [Figure 9] shows the performance metrics across the models for each type of dataset (from 17 features, down to 6). Note that the original transformed dataset performed better than the reduced dimensionality datasets in an indication that additional information was needed to improve model performance. Nevertheless, the overall performance of models was below what was anticipated and desired.

Transformed Dataset (17 features)

Metric/Model	LR	DT	RF	GB	SVM
Accuracy	0.578947	0.526316	0.473684	0.473684	0.473684
F1-Score	0.592593	0.592593	0.500000	0.642857	0.592593
AUC	0.572222	0.544444	0.500000	0.500000	0.500000
Sensitivity	0.444444	0.888889	1.000000	1.000000	1.000000
Specificity	0.571429	0.500000	0.473684	0.473684	0.473684

Reduced Dimensionality Set (9 features)

Metric/Model	LR	DT	RF	GB	SVM
Accuracy	0.421053	0.421053	0.368421	0.473684	0.421053
F1-Score	0.592593	0.592593	0.500000	0.642857	0.592593
AUC	0.444444	0.444444	0.383333	0.500000	0.444444
Sensitivity	0.888889	0.888889	0.666667	1.000000	0.888889
Specificity	0.444444	0.444444	0.400000	0.473684	0.444444

Further Reduced Dimensionality Set (6 features)

Metric/Model	LR	DT	RF	GB	SVM
Accuracy	0.473684	0.473684	0.473684	0.473684	0.473684
F1-Score	0.592593	0.592593	0.500000	0.642857	0.592593
AUC	0.500000	0.500000	0.500000	0.500000	0.500000
Sensitivity	1.000000	1.000000	1.000000	1.000000	1.000000
Specificity	0.473684	0.473684	0.473684	0.473684	0.473684

Figure 9 Performance Metrics Across ML Models and for Different Datasets Analyzed

The Turning Point

Despite extensive tuning (Grid Search Cross Validation with 50+ hyperparameters) and imbalance handling (SMOTE, weighted loss function), no model surpassed 64% F1. The confusion matrix for the best performing model [Figure 10] told the sobering truth - the best model misclassified 48% (9/19) of at-risk students.

```
plt.title('Confusion Matrix for LR')
plt.show()
```

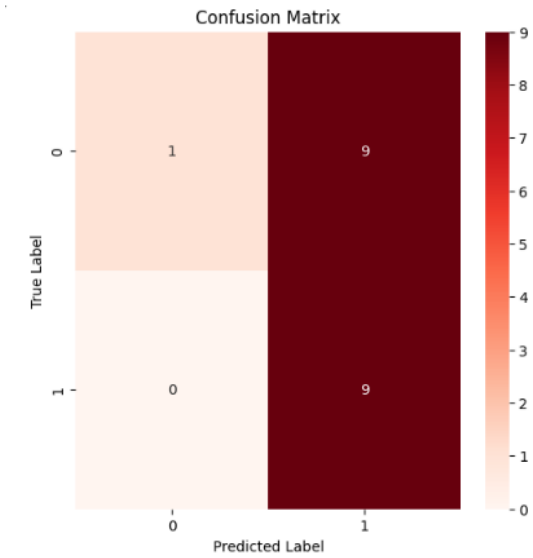

Figure 10 Confusion Matrix for LR w/Weighted Loss Function & SMOTE Oversampling

The GPA Threshold Revelation

As models faltered, GPA consistently emerged as the North Star:

- PCA Loadings: 0.66 (PC4) [Figure 2] and
- Logistic Regression: coefficient of 0.79:

```
Accuracy: 0.5263157894736842
F1 Score: 0.6666666666666666
AUC: 0.55
Sensitivity: 1.0
Specificity: 0.5

Coefficients:
 Age_scaled: -0.2351
 Sex: 0.8474
 White: -0.4005
 Asian: 0.4048
 AmIndian: 0.2241
 PacIslander: 0.3600
 Black: -0.0362
 Hispanic: 0.0212
 ELL_T: 0.7655
 ESE_T: 0.3203
 504_T: -0.6930
 FST_Reading_Level_Scaled: -0.1510
 Math_Score_Scaled: 0.1692
 Unweighted_GPA_Scaled: 0.7906
 Grade_Scaled: -0.0343
 Days_Absent_YTD_Scaled: -0.1049
 Referrals_YTD_Scaled: 0.3823
```

Figure 11 Coefficients from Logistic Regression Model with Best Performance

The Decision Tree model showed that with GPA≥2.88 is a primary decision-making point:

```
Accuracy: 0.5263157894736842
F1 Score: 0.64
AUC: 0.5444444444444444
Sensitivity: 0.8888888888888888
Specificity: 0.5
```

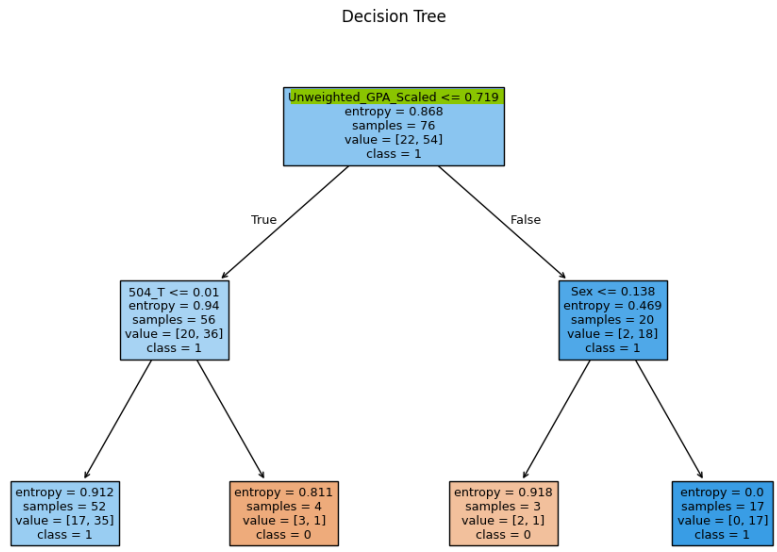

Figure 12 Decision Tree Shows GPA >= 2.88 (Scaled: >= 0.719) as Root Decision Point

And GPA threshold analysis [Figure 5] pinpointed to a GPA of 2.5 as optimal balance:

- At or above 2.5: nearly 60% success rate,
- Below 2.5: nearly 80% unsuccessful rate
- And given the decision tree rule of a GPA >= 2.88 [Figure 12], I opted for 2.5 GPA as a guideline for a decision-making point, which also matches observations from administration and counselors

Next, we will discuss these results and conclude with an operational recommendation to satisfy the key question in this study.

Discussion of Results

This research set out to answer a central question: How can we predict student success in a high school cybersecurity pathway using school records? The journey through data transformation, dimensionality reduction, and model selection revealed both the promise and the limitations of educational data mining in a real-world school context.

Key Findings and Their Implications

From the outset, the data transformation process highlighted that certain features-most notably unweighted GPA, reading scores, and indicators of English language learning (ELL) or exceptional student education (ESE) status-were strongly associated with student outcomes. As shown in the initial descriptive statistics and correlation heatmaps (Appendix D), students with an unweighted GPA above 2.5 were far more likely to succeed, while ELL students and those with non-gifted ESE status faced greater challenges. These findings echo previous research emphasizing the predictive power of academic records and demographic factors (Batool et al., 2023; Costa-Mendes et al., 2021; Dustegor, 2020).

The process of reducing dimensionality through correlation analysis and principal component analysis (PCA) was instructive. While the literature often suggests that simplification can improve model performance (Costa et al., 2017), in this study, reducing features from 17 to 6 led to a decline in predictive accuracy and F1 score. This outcome highlights the nuanced nature of educational datasets: what appears redundant statistically may, in fact, capture critical context about student readiness.

Model Performance and Selection Criteria

Five machine learning models-Logistic Regression, Decision Tree, Random Forest, Gradient Boost, and SVM-were rigorously tested, each with hyperparameter tuning and cross-validation. The results, summarized in [Figure 9] revealed that none of the models achieved high predictive power, with F1 scores ranging from 0.50 to 0.64 and AUC values hovering near 0.55. Even after addressing class imbalance with weighted loss functions and SMOTE oversampling (Wu, 2025), improvements were marginal.

Despite these limitations, one insight was consistent across all models: unweighted GPA was the most influential predictor. The decision tree model, for instance, split at a GPA threshold of approximately 2.88, and logistic regression assigned

the highest coefficient to GPA. Further analysis revealed that a threshold of 2.5 maximized the correct identification of both successful and unsuccessful students, that matched school counselors' observations.

Implications for Cybersecurity Education

These findings suggest that, for the current dataset and context, a simple rule based on GPA can be as effective as more complex machine learning models. This aligns with the broader literature, which cautions against overcomplicating predictive analytics in small or imbalanced educational datasets (Ahmed, 2024; Al-Bahri & Al Mayahi, 2020). Implementing a 2.5 GPA threshold as a guideline for Cyber 1 admissions, with targeted remediation for students below this mark, offers a transparent and actionable approach for schools seeking to improve student outcomes in technical pathways.

Limitations

Several limitations temper these conclusions. The sample size (n=95) restricts the generalizability and statistical power of the models, a challenge noted in similar educational data mining studies (Kotsiantis et al., 2004). The absence of psychosocial survey data-due to technical delays-means that important predictors such as motivation, self-efficacy, and career interest were not included in this phase. Additionally, while GPA is a strong predictor, it may inadvertently disadvantage certain groups, such as ELL students, if used as the sole criterion, hence our suggestion to use it as a guideline.

Suggestions for Future Research

Future research should prioritize the integration of validated psychosocial survey data (Maphosa et al., 2022) to capture a fuller picture of student readiness. Expanding the cohort size and conducting longitudinal studies will allow for more robust modeling and the exploration of more sophisticated algorithms, such as ensemble or deep learning methods, as recommended in the literature (Bachmann, 2021). Regular equity audits should also be implemented to ensure that selection criteria do not unintentionally reinforce existing disparities.

In our data transformation analysis, we found that females (41%) are substantially less successful than males (72%) even though those unsuccessful females possessed high reading levels (See Appendix C), as well as high math scores as seen in [Figure 13]. I hope this may be elucidated with the psychosocial survey data planned for the

continuation of this effort and it certainly deserves further study as this is an important population for the future of the Cybersecurity workforce.

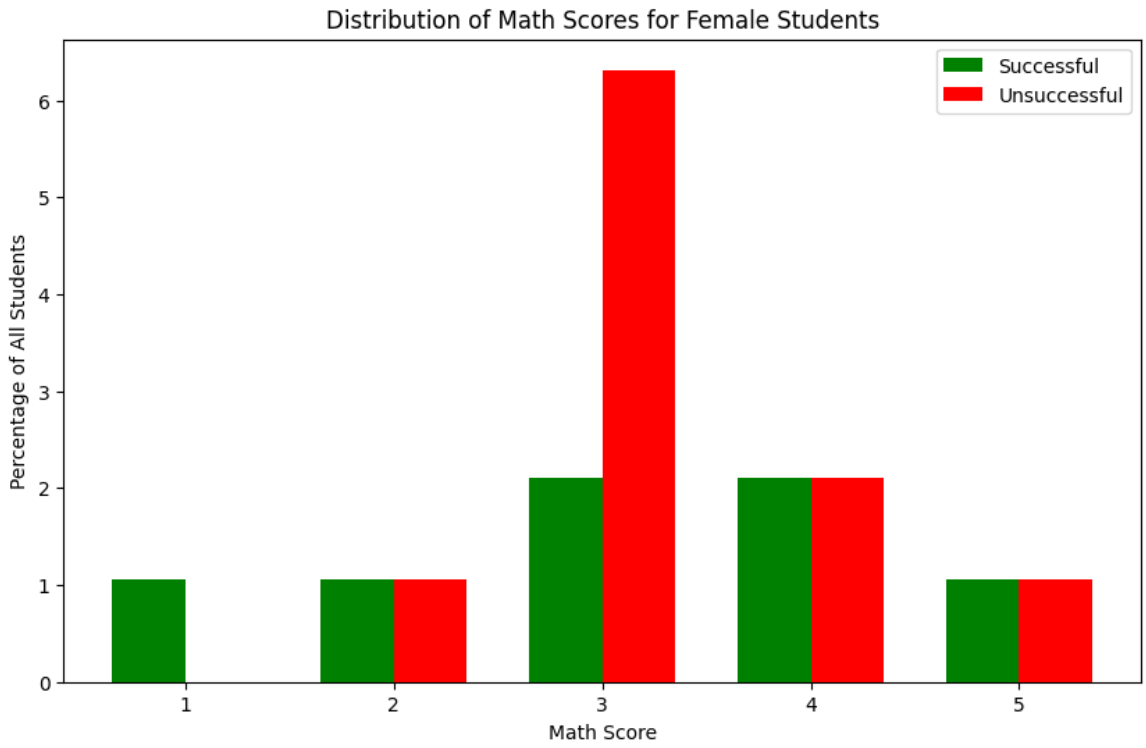

Figure 13 Math Scores for Successful and Unsuccessful Females

Conclusion

This study set out to use educational data mining and machine learning to predict which students are most likely to succeed in a high school cybersecurity pathway, relying solely on school records. The original question was: How can we predict student success in a high school cybersecurity pathway using school records? The process demonstrated both the power and the limitations of data-driven decision-making in education.

The most consistent and actionable finding was the central role of unweighted GPA in predicting student success. Despite experimenting with a range of sophisticated machine learning models, none substantially outperformed a simple GPA threshold. A cutoff of 2.5 was found to balance the identification of students likely to succeed with the practical needs of school counselors and administrators. This result is consistent with prior research highlighting the predictive strength of academic performance measures in STEM and technical education (Costa-Mendes et al., 2021; Geiser & Santelices, 2007).

While the models themselves were not highly accurate-likely due to the modest dataset size and the absence of psychosocial features, the process yielded important operational insights. It reinforced the importance of transparency and interpretability in predictive analytics for educational settings, as well as the need for ongoing collaboration between data scientists and educators.

Looking ahead, the next phase of this research will involve integrating survey-based measures of student motivation, persistence, and interest, as well as expanding the dataset to include future cohorts, and further analysis of the reasons for female students lower performance. These steps are essential for building more nuanced, equitable, and effective predictive models. Ultimately, the journey from raw data to actionable insights in this project underscores a broader lesson: data science can illuminate pathways to student success, but it must always be paired with professional judgment and a commitment to educational equity.

References

- Ahmed, E. (2024). Student performance prediction using machine learning algorithms. Applied Computational Intelligence and Soft Computing, 2024. https://doi.org/10.1155/2024/4067721
- Al-Bahri, M., Al Mayahi, K. (2020) Machine Learning Based Predicting Student Academic Success. Retrieved March 27, 2025. 12th International Congress on Ultra Modern Telecommunications and Control Systems Workshop (ICUMT) https://bit.ly/3RpOJZW
- Bachmann, Janio. "Predicting Grades for the School Year." Kaggle, 2021, www.kaggle.com/code/janiobachmann/predicting-grades-for-the-school-year
- Batool, S., Rashid, J., Nisar, M.W. et al. Educational data mining to predict students' academic performance: A survey study. Educ Inf Technol 28, 905–971 (2023). Retrieved Mar 27, 2025 https://doi.org/10.1007/s10639-022-11152-y
- "Classification Methods: Problems and Solutions." CognitiveClass.ai, https://apps.cognitiveclass.ai/learning/course/course-v1:IBMSkillsNetwork+AI0120EN+v1/home Accessed 22 Mar. 2025.
- "Data Analysis with Python." CognitiveClass.ai, use this link Accessed 22 Mar. 2025.
- "Data Science Methodology." CognitiveClass.ai, use this link. Accessed 22 Mar. 2025.
- Chada, A. (2020) “Evaluation Metrics” Distilled AI, accessed April 27, 2025 https://aman.ai/primers/ai/evaluation-metrics/#sensitivity-and-specificity
- Costa, E.B., Fonseca, B., Santana, M.A., de Araújo, F.F., & Rego, J. (2017) "Evaluating the effectiveness of educational data mining techniques for early prediction of students' academic failure." Computers in Human Behavior, vol. 73, Aug. 2017, pp. 247–256. ScienceDirect, accessed 27 Mar 2025 https://www.sciencedirect.com/science/article/abs/pii/S0747563217300596
- Costa-Mendes, R., Oliveira, T., Castelli, M., & Cruz-Jesus, F. (2021). A machine learning approximation of the 2015 Portuguese high school student grades: A hybrid approach. Education and Information Technologies, 26,

1527-1547. Retrieved on 27 Mar 2025
https://link.springer.com/article/10.1007/s10639-020-10316-y

- DeJong, B. P. and Karadogan, E. (2024), "Using Machine Learning to Predict Student Success in Undergraduate Engineering Programs," 2024 IEEE 3rd International Conference on Computing and Machine Intelligence (ICMI), Mt Pleasant, MI, USA, 2024, pp. 1-5. Accessed on 28 Mar 2025 https://ieeexplore.ieee.org/document/10585656?figureId=fig2#fig2
- Dustegor, D. (2020) Predicting Academic Success in Higher Education Literature Review and Best Practices. International Journal of Education Technology in Higher Education (17 (1)) Retrieved from ResearchGate on March 27, 2025 https://www.researchgate.net/publication/338717237_Predicting_Academic_Success_in_Higher_Education_Literature_Review_and_Best_Practices
- Geiser, S., & Santelices, M. V. (2007). Validity of high-school grades in predicting student success beyond the freshman year: High-school record vs. standardized tests as indicators of four-year college outcomes. Research & Occasional Paper Series: CSHE.6.07. Center for Studies in Higher Education. Retrieved from ERIC on April 27, 2025 https://files.eric.ed.gov/fulltext/ED502858.pdf
- Kotsiantis, S.B., Pierrakeas, C.J., & Pintelas, P.E. "Predicting Students' Performance in Distance Learning Using Machine Learning Techniques." Applied Artificial Intelligence, vol. 18, no. 5, May 2004, pp. 411–426. Taylor & Francis Online, www.tandfonline.com/doi/abs/10.1080/08839510490442058 Accessed 9 Mar 2025.
- M. Maphosa, W. Doorsamy and B. S. Paul (2022). "Factors Influencing Students' Choice of and Success in STEM: A Bibliometric Analysis and Topic Modeling Approach," in IEEE Transactions on Education, vol. 65, no. 4, pp. 657-669, Nov. 2022. Accessed on 28 Mar 2025 https://ieeexplore.ieee.org/document/9741836
- "Predictive Modeling Fundamentals I." CognitiveClass.ai, use this link. Accessed 22 Mar. 2025.
- "Prompt Engineering for Everyone." CognitiveClass.ai, use this link. Accessed 22 Mar. 2025.

- Tkáčik, K. (2023). Predicting student success in cybersecurity training [Master's thesis, Masaryk University]. https://is.muni.cz/th/dkm2u/?lang=en
- Wu, N. (2025) Presentation on How to Address Analysis of Imbalanced Data via online lecture and presentation https://colab.research.google.com/drive/1pAUQVFy_oiw6fobv6Z1LaWY4m2TBIird?usp=sharing accessed on 4/22/2025

Appendix A – School-provided Data

The data provided by the school's SIS were these 27 features:

- Student ID
- Last First M (Student Name)
- Grade
- Birthdate
- Age (as of 3/21/25)
- Sex
- Unweighted GPA
- Race: White
- Race: Asian
- Race: American Indian or Alaska Native
- Race: Native Hawaiian or Other Pacific Islander
- Race: Black or African American
- Ethnicity: Hispanic or Latino
- Single Ethnicity
- English Language Learner
- ESE Primary Exceptionality
- 504 Plan
- FST Reading Overall Score Administration Date
- FST Reading Overall Score Achievement Level Score
- Graduation Test Algebra I EOC or Concordant Pass Date or Waiver

- Graduation Test Algebra I EOC or Concordant Passed or Waiver (Met Algebra 1 Requirement)
- BST Algebra 1 EOC - Overall Scale Score Administration Date
- BST Algebra 1 EOC - Overall Scale Score Achievement Level Score
- FST Mathematics Overall Score Administration Date
- FST Mathematics Overall Score Achievement Level Score
- Days Absent YTD
- Number of Referrals

Appendix B – Survey Description

Our school records were limited to identifiers (name, ID) that were later anonymized for this analysis, demographics, GPA, attendance, disciplinary data, and standardized test scores. As not all data was readily available from our school records, I needed to collect them directly from the student via survey. Based on research (Bachmann, 2020; Dustegor, 2020; Costa-Mendes, 2021; Maphosa, 2022), I was able to identify nearly 32 features that could be used in this analysis.

Upon further review, I saw that many of them would not be available from my main data sources (from my School or District), and that some may be too sensitive to request via survey, such as income levels or parents' highest level of education achieved.

Based on Maphosa et al.'s (2022) research of over 179 articles related to why students choose and succeed in STEM, I decided to focus on the features I felt would be most impactful, such as math ability, cybersecurity career interest, school experience, persistence, achievement, and self-efficacy (Maphosa et al., 2022), reducing to 14 features in the raw data obtained from my school and deciding to gather 14 more features from student surveys:

- Reason for enrolling in Cybersecurity program
- Previous knowledge of IT (computer, programming, networking)
- Confidence level in learning IT
- STEM participation (Clubs)
- Ability to think critically (solve problems) and persist
- Daily time dedicated to studying, game-playing, commuting to school
- Preferred learning method (visual, auditory, etc.)
- Reading skills
- Computer and internet access at home
- Other features

Consent forms were signed so students and parents could provide information.

It is important to note that, due to erroneous technical settings, the surveys did not register associated with the student, which was needed to match the school records,

as originally intended. Although I was able to get the data back, it was too close to the deadline for this report and thus moved to future work in this topic.

The draft survey can be seen at https://forms.office.com/r/vXYRV6sGdF

Appendix C – Extract, Transform and Load (ETL) of School Data

See Google Colab notebook via link:

https://colab.research.google.com/drive/1LzBlDWBsZwkAnMfvZ9UH5LWUV02Bpzzq?usp=sharing

Appendix D – Analyzing Cybersecurity Student Data via ML Models

See below Google Colab notebook via link:

https://colab.research.google.com/drive/15IEQgWa0vGugmcaxf1qF9w38clsMe6-y?usp=sharing

Appendix E - Project Timeline and Hours Breakdown

Week/Day	Hours	Activity/ies
1/27-2/2/25	4	Researched and defined data elements (features) for the original Math Covdi-19 Gap analysis project
2/3-9/25	4	Worked on obtaining School District IRB approval for the data needed for the above/original project
2/10-16/25	2	As IRB route for the original project did not come through, pivoted to the idea of predicting student success in Cybersecurity program
2/17-23/25	3	Defined data elements (features) needed for this new question
2/24-3/2/25	4	Worked with school staff to obtain the (majority) of above elements from school systems and records for current and dropped Cybersecurity students
3/3-9/25	12	Reviewed 24 different literature articles and selecting those relevant to this research
3/10-16/25	6	Learned about different classification models (Decision Tree, Linear Regression, SVM, KNN, etc) from CognitiveClass.ai course
3/17-23/25	8	Started writing report and started on Google colab notebook
3/24-30/25	12	Reviewed school-provided data, transformed it in Google Collab, and checked transformations as Data Pre-processing ste. Updated report/paper
3/31-4/6/25	9	Worked on incorporating advisor comments including data visualization
4/7-13/25	4	Created data visualizations of the dataset to gain further insights
4/14-19/25	3	Conducted Primary Component Analysis (PCA) together with Correlation and Variance Info Analysis to reduce dimensionality of the data.

4/20/25	3	Analyzed different ML models (Logistic Regression, Decision Tree, Random Forest, Gradient Boost, SVM) to predict Cybersecurity Readiness
4/21/25	6	Worked on issues with models not achieving reasonable performance metrics (accuracy, F1, AUC) - studying modifications to create weighted balance function to the moderate imbalance (70/30) of the data and create SMOTE oversampling to seek to improve metrics although they did not seem to work that well.
4/24/25	2	Reviewed lecture and materials from Dr. Wu's and Dr. Pierce's lecture on imbalanced data management
4/26/25	6	Reviewed data and machine learning models, worked with different hyperparameter values and Grid Search optimization, improved visualization of results in Google Colab, and Derived simpler rule from analysis of Unweighted GPA and the target (Succesful).
4/27/25	2	Computer would not turn on. Had to get another one. Decided to concentrate on my data science journey in this project as the main theme in my report and 3-minute video. Finished drafting report and started drafting video script.
4/28/25	3	Reworked some of the visualization on the ETL Google Colab to reveal new findings especially for females and Hispanic students
4/29/25	3	Finished the above rework, refined the report, included new figures because of above rework, and revised 3-minute video script.
4/30/25	2	Read and revised report
5/1-2/25	4	Scripted, created slide and created 3-minute video (after many tries)
Total Hours	**102**	